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**RELIEFLINK: A BLOCKCHAIN-ENABLED MOBILE AND WEB DONATION
MANAGEMENT SYSTEM WITH PRESCRIPTIVE ANALYTICS**

A Capstone Project

Submitted to the Faculty of

National University - College of Computing and Information Technologies

In Partial Fulfillment of the Requirements for the Degree

Bachelor of Science in Information Technology with

Specialization in Mobile and Web Application

BY:

Arillo, Francis Louis M.

Estrella, Rhyza Ann H.

Rivera, Shirene D.

Romero, Cyrell Jane D.

Ser, Laurence

Edward Matthew T. Sanmocte

Capstone Adviser

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RECOMMENDATION FOR ORAL EXAMINATION

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CHAPTER 1

INTRODUCTION

Charitable organizations play a crucial role in addressing social issues by providing financial assistance, relief goods, and essential services to underserved communities [1]. These organizations rely heavily on donations to support programs related to education, health, disaster relief, and community development. However, despite the positive impact of these initiatives, donation management particularly for cash-based contributions continues to face challenges related to transparency, accountability, and trust [2].

In traditional donation systems, cash contributions are commonly recorded using manual documentation or centralized databases. These methods are prone to human error, delayed reporting, data manipulation, and limited auditability [3]. Donors often have little to no visibility into how their contributions are distributed and utilized, while partner organizations must rely on internal reporting mechanisms that may lack real-time verification. This gap creates uncertainty and can weaken donor confidence, especially when proof of fund utilization is not consistently required or validated.

Blockchain technology offers a promising solution to these challenges by providing an immutable, transparent, and auditable ledger for recording transactions [4]. While existing blockchain-based donation platforms have improved transparency, most are designed for cryptocurrency-based donations, making them unsuitable for organizations that primarily handle cash contributions. Furthermore, many current systems do not require mandatory proof of fund usage, limiting their ability to fully verify whether donations are used as intended.

To address these gaps, this study proposes ReliefLink: A Blockchain-Enabled Mobile and Web Donation Management System with Prescriptive Analytics. The system is designed to record all cash donations and fund disbursements on a blockchain while integrating off-chain receipt verification to ensure that every transaction is supported by valid documentation. By combining on-chain transaction immutability with off-chain proof validation, ReliefLink enhances transparency and accountability throughout the entire donation lifecycle, from donor contribution to final fund utilization [5].

In addition to transparency, the system incorporates prescriptive analytics to support data-driven decision-making. By analyzing donation patterns, fund usage, and

beneficiary needs, the system provides actionable recommendations that help organizations optimize resource allocation and improve program effectiveness. This feature addresses common needs of beneficiaries, such as timely assistance, fair distribution of resources, and responsiveness to urgent community needs.

Overall, ReliefLink aims to strengthen donor trust, improve operational efficiency, and ensure responsible fund management by offering a secure, transparent, and intelligent donation management platform tailored for cash-based charitable organizations.

1.1 PROJECT CONTEXT

The Keh Family Foundation, the corporate social responsibility arm of McBride Corporation, is a God-centered nonprofit organization committed to improving the lives of people around them, beginning with McBride employees and their families. In line with their mission, the foundation provides medical and educational assistance, implements programs that benefit surrounding communities, and collaborates with other organizations to support disadvantaged groups. Despite their dedication, the foundation faces challenges in managing donations efficiently, maintaining transparency, and ensuring accountability, especially as their outreach and programs continue to grow. Manual tracking and fragmented systems limit real-time visibility of contributions and the proper allocation of resources, which can affect donor confidence and hinder timely support to beneficiaries.

To address these challenges, this project proposes RELIEFLINK, a blockchain-enabled mobile and web donation management system integrated with prescriptive analytics. The platform ensures secure, traceable, and tamper-proof donation records, allowing donors and administrators to verify transactions with confidence. Its mobile and web interfaces streamline donation collection, monitoring, and reporting, while prescriptive analytics provides actionable recommendations for optimal fund allocation based on historical trends and community needs. By combining transparency, efficiency, and data-driven insights, RELIEFLINK empowers the Keh Family Foundation to enhance its programs, strengthen trust with stakeholders, and maximize its impact on the communities it serves.

1.2 OBJECTIVES

General Objective

To develop a blockchain-enabled mobile and web-based donation management system with prescriptive analytics that provides a transparent, accountable, and reliable method for managing cash-based donations.

Specific Objectives

1. To develop a web-based donation management system that securely records and monitors cash-based donations and fund disbursements using blockchain technology.
2. To develop a mobile application that allows stakeholders to access real-time donation records and distribution status.
3. To implement a blockchain technology with prescriptive analytics, enabling mandatory receipt and document verification while generating data-driven recommendations to improve fund utilization and allocation decisions.
4. To evaluate the system's quality based on ISO/IEC 25010 software quality standards, focusing on:
 - Functionality,
 - Reliability,
 - Usability,
 - Security,
 - and Maintainability.

1.3 SCOPE AND DELIMITATION

Scope of the Project

Both Android and iOS devices are supported by the mobile application, which requires an active internet connection and is intended exclusively for users or donors. Through this platform, donors can register, log in, make cash-based donations, and

view their donation history anytime and anywhere, provided that internet access is available, ensuring convenience, accessibility, and real-time interaction via mobile devices. Complementing the mobile application, the web-based component of the Donation Management System is designed exclusively for authorized Keh Family Foundation administrators and staff, providing controlled and secure access to organizational data. It implements role-based access control, assigning specific permissions and functions based on user responsibilities, and focuses on administrative operations such as managing cash-based donation records, monitoring and verifying transactions, generating reports, and overseeing donor information. As a web application, it operates through a wired internet connection within the foundation's work environment, supporting stable and centralized system access while promoting accountability, data integrity, and efficient management of donation information across the organization

Delimitation of the Project

This project is limited to the acceptance and recording of cash-based donations only, thereby restricting the system's functionality exclusively to donation-related activities. It does not accommodate any other type of financial transaction, including payments for goods or services, refunds, fund transfers, disbursements, or withdrawals. In addition, the system does not support direct monetary exchanges between users, automated fund distribution to beneficiaries, or integration with external banking institutions, digital wallets, or online payment gateways. As a result, its capabilities are confined to the documentation, tracking, verification, and reporting of donation records. This delimitation ensures that the system maintains a clear focus on enhancing transparency, accountability, and proper management of donation data within the defined scope of the study.

CHAPTER 2

REVIEW OF RELATED LITERATURE, STUDIES AND SYSTEMS

2.1 RELATED LITERATURES/STUDIES

The growing demand for transparency and accountability in philanthropy has catalyzed the exploration of technological solutions. Blockchain technology, with its core attributes of decentralization, immutability, and cryptographic verifiability, presents a transformative framework for rebuilding donor trust. This review of related literature is structured into two key sections to provide a comprehensive foundation for the proposed donation system. First, it analyzes existing related systems, operational platforms and applications that demonstrate practical implementations of blockchain for social good. This analysis reveals current capabilities and identifies functional gaps, particularly regarding cash-based donations and mandatory proof of usage. Second, it examines related studies, academic and theoretical works that provide the economic rationale, technical validation, and critical discourse surrounding the application of distributed ledger technology to charitable giving. Together, this dual analysis establishes the context, justifies the technological approach, and highlights the specific research gap this capstone project aims to address. The development of the aforementioned systems is informed and critiqued by a growing body of academic research, with [1] providing the foundational economic rationale for blockchain by explaining how the technology drastically reduces the costs of verification and networking, offering a core theoretical argument for its use in creating low cost, high trust environments for transactions such as charitable donations. Study [2] provides a foundational and comprehensive overview of blockchain technology by detailing the core technical components such as cryptography, consensus mechanisms, and data structures that enable decentralization, immutability, and transparency, while tracing the technology's evolution from its cryptocurrency origins to its disruptive potential across sectors including finance, supply chain, and healthcare, thereby establishing the technological bedrock upon which transparent donation platforms are built. Study [3] presents a systematic review of blockchain applications in the energy sector, with findings on transparent and decentralized data management that are directly transferable to philanthropic systems by demonstrating how immutable and auditable records support accountability and traceability in complex multi-party transactions and resource flows, which are essential for verifying the end to end journey of charitable funds. Study [4] develops a detailed taxonomy of blockchain technologies by classifying systems according to architectural and governance principles, providing a

crucial framework for understanding design choices such as public or private ledgers and consensus mechanisms that are vital for making informed technical decisions in the development of a transparent donation system. Study [5] offers a systematic bibliometric analysis of blockchain integration into education by mapping academic discourse from 2018 to 2022, identifying transformative potentials such as securing records and credential verification alongside adoption hurdles, and presenting a structured framework for understanding how decentralization and immutability are translated into non-financial sectors, offering a valuable conceptual model for enhancing transparency and trust in philanthropic systems.

2.2 RELATED SYSTEMS

Table 1 Gap Analysis Table of Related Systems and Applications

Feature(s)	ReliefLink	GiveTrack (BitGive)	Binance Charity	Giveth	WFP Building Blocks	Gitcoins Grants
<i>Country</i>	Philippines	International	International	International	International (Humanitarian)	International
<i>Application Type</i>	Web & Mobile	Web-Based	Web & Mobile	Web-Based (Dapp)	Enterprise System	Web-Based
Blockchain-Based Ledger	✓	✓	✓	✓	✓	✓
Cryptocurrency Donations	✓	✓	✓	✓	✓	✓
Cash-Based Donations	✓				✓	
Public Transaction Transparency	✓	✓	✓	✓		✓
Smart Contract Utilization	✓	✓	✓	✓	✓	✓
Donor Fund Tracking	✓	✓	✓	✓	✓	✓
Mandatory Proof -of Usage Submission	✓					
Automated Verification of Fund Usage	✓					
Evidence-Based Validation (Receipts, Media, Reports)	✓					
Community or System Based Oversight	✓			✓	✓	✓
Immutable Recording After Verification	✓					

Table 1 illustrates the comparison of several platforms, which shows the key features of a Donation Management System, such as Blockchain-Based Ledger, Cryptocurrency Donations, Cash-Based Donations, Public Transaction Transparency, Smart Contract Utilization, Donor Fund Tracking, Mandatory Proof -of Usage Submission, Automated Verification of Fund Usage, Evidence-Based Validation (Receipts, Media, Reports), Community or System Based Oversight, and Immutable Recording After Verification

GiveTrack

GiveTrack by BitGive is recognized as an early platform dedicated to transparent cryptocurrency philanthropy. It allows donors to trace the flow of their donations from contribution to final project expenditure on a public ledger, providing a foundational model for transactional visibility in charitable funding.

Binance Charity

Binance Charity the philanthropic arm of the Binance cryptocurrency exchange, executes initiatives like "Lunch for Children" using its native blockchain. It records all donations and disbursements on-chain, creating an immutable and publicly auditable record for large-scale humanitarian campaigns, demonstrating scalability in blockchain-based aid.

Giveth

Giveth represents a community-driven, decentralized approach. As an open-source Ethereum application, it facilitates traceable donations for social good projects and incorporates community verification, exploring a model of decentralized oversight and governance within philanthropic funding.

WFP Building Blocks

World Food Programme's Building Blocks is a seminal enterprise-level implementation in the humanitarian sector. This system uses a permissioned blockchain to manage cash-based assistance for refugees, significantly reducing transaction costs and leakage. Its success provides a critical case study for applying blockchain to track the distribution of traditional fiat-based aid securely and efficiently.

Gitcoin

Gitcoin Grants has pioneered the use of blockchain for funding public goods through quadratic funding. While focused on the open-source software ecosystem, its model—where all contributions and matching fund distributions are transparently recorded and calculated on-chain—offers a powerful framework for democratizing and verifying community-driven donation matching.

2.3 SYNTHESIS

The review of existing systems and scholarly literature reveals a consistent trajectory: blockchain technology is effectively used to create transparent, immutable records of philanthropic transactions, yet a significant gap remains between recording a transaction and verifying its real-world impact. Operational platforms like GiveTrack and Binance Charity successfully leverage public ledgers to provide donors with visibility into the flow of funds, primarily for cryptocurrency donations. However, these systems generally treat the confirmation that funds were used as intended as a separate, often manual reporting process, rather than a mandatory, automated checkpoint within the blockchain protocol itself. This reflects the broader challenge identified in academic studies—often termed the "oracle problem"—of reliably connecting on-chain certainty with off-chain truth.

This analysis highlights a clear opportunity for innovation. While current solutions excel at transactional transparency, they do not architecturally enforce verified proof of proper fund usage as a condition for finalizing a donation record, especially for cash-based contributions. Therefore, this project addresses this gap by proposing a system that integrates mandatory, evidence-based verification into the core transaction flow. By designing a process where the immutable recording of a successful donation on the blockchain is contingent upon the submission and validation of proof-of-usage, the proposed model aims to advance from passive transparency to active, verifiable accountability.

CHAPTER 3

TECHNICAL BACKGROUND

This chapter presents the detailed technologies to be used in the development of the proposed system. Also, a through discussion on the technical description of the proposed system and its components are described.

3.1 DETAILS OF THE TECHNOLOGY TO BE USED

Software/Hardware Requirements

Table 2 Software Requirements for Systems Development

Software Classification	Software Name
Integrated Development Environment (IDE)	Visual Studio Code
Programming Languages and Frameworks	Flutter & Dart, MERN Stack (MongoDB, Express.js, Node.js)
Database	MongoDB Atlas
Third-Party Libraries	Render / Railway Stripe API Infura API (Polygon PoS integration) Twilio API
Others	Figma Git Canva

Table 2 presents the required software components essential for the development of the ReliefLink mobile and web applications. Detailed below are the distinct software entities, elucidating their respective usage and applications within the project framework.

Integrated Development Environment (IDE)

Visual Studio Code

Visual Studio Code is a development environment used for building the system. Its lightweight design, versatility, and support for multiple languages and extensions allow developers to write and organize code, debug errors, manage version control with Git, and integrate libraries and APIs for implementing features such as blockchain and analytics.

Programming Languages and Frameworks

Flutter & Dart

Flutter enables the creation of cross-platform apps from a single codebase, while Dart provides the programming framework for building fast, efficient, and responsive interfaces. Together, they allow developers to design user-friendly interfaces, implement system functionalities, and ensure consistent performance across Android, iOS, and web platforms.

MERN Stack (MongoDB, Express.js, Node.js)

MongoDB stores and manages donation and user data, Express.js handles server-side routing and requests, and Node.js provides a runtime environment for executing JavaScript code. Together, they enable efficient data management, seamless communication between the server and client, and reliable system performance for web functionalities.

Database

MongoDB Atlas

MongoDB Atlas is a cloud-based database used to store and manage application data. It provides secure, scalable, and accessible storage for donation records, user information, and transaction data, ensuring reliable performance and real-time access from both web and mobile platforms.

Third-Party Libraries

Render / Railway

Render / Railway are cloud-based platforms used for deploying and hosting web applications. They provide an environment for running backend servers and services, ensuring continuous availability, scalability, and real-time access for users while simplifying deployment and management of application resources.

Stripe API

Stripe API is a payment processing platform used to handle online financial transactions. It enables secure and reliable processing of cash-based donations, manages transaction data, and provides real-time verification while ensuring the safety and integrity of donor information.

Infura API (Polygon PoS integration)

Infura API is a blockchain infrastructure service used to connect applications to the Polygon Proof-of-Stake (PoS) network. It enables secure and reliable interaction with the blockchain, allowing the system to record, verify, and track transactions in a decentralized and tamper-resistant manner.

Twilio API

Twilio API is a cloud communication platform used to send notifications and alerts to users. It enables the system to deliver real-time messages, such as donation confirmations or updates, through SMS or other messaging channels, ensuring timely communication and engagement with users.

Others

Figma

Figma is a web-based design tool used for creating user interface and experience designs. It allows developers and designers to prototype screens, design layouts, and visualize application workflows, ensuring a user-friendly and consistent interface across both mobile and web platforms.

Canva

Canva is a web-based graphic design tool used to create visual assets for applications. It allows designers to produce backgrounds, icons, and other interface elements, ensuring visually appealing and consistent designs for both mobile and web platforms.

Git

Git is a version control system used to track and manage changes in the application’s source code. It allows developers to maintain code history, collaborate efficiently, and manage updates or revisions across different stages of development.

Table 3 Hardware Requirements for Systems Development

Hardware Component	Technical Specifications
Computer Desktop or Laptop	The system requires a minimum of an Intel Core i5 processor or its equivalent, with 8 GB of RAM and at least 150 GB of available hard disk space or 120 GB of solid-state drive storage.
Smartphone	Supported mobile platforms include Android OS version 13.0 or higher with a minimum of 8 GB RAM, and iOS version 17.0 or above.
Network	An internet connection is required for both the mobile and web applications to upload and access organizational data.

Table 3 presents the hardware requirements along with their respective specifications, utilized in the development, implementation, and testing of the ReliefLink mobile and web applications.

Network Requirements

Having a stable and reliable internet connection is necessary to ensure that the systems mobile and web-based donation management system is working properly. This is because the donation management system is using a blockchain to ensure that donations are safe and that data is up-to-date in real-time worldwide. This means that a reliable internet connection is necessary. The network connection will allow users to utilize the system capabilities for donation submittal, transaction validation, creating reports, and analyzing data. The connection will further help the administrators in analyzing the donations and validating the blockchain information.

Wi-Fi (Wireless Fidelity) / Wireless Connection

Users and administrators can access the system via a wireless internet connection. It is recommended to have at least 10 Mbps internet speed for seamless performance, swift data synchronization, and real-time blockchain validation to ensure uninterrupted descriptive and prescriptive analytics dashboards.

Wired Connection

For the office-based administrators and the representatives of NGOs, a wired internet connection should be preferred in order to create a more stable and secure networking environment. At least an internet speed of 10 Mbps is required in order for the systems' functionalities to be maximized, especially in the management of larger sets of data, creating reports, and tracking transactions of donations received.

Mobile Data

Donors, relief workers, and volunteers can all use the system Mobile App with their mobile data. As long as you've got at least a 5 Mbps connection, everything runs smoothly and transactions go through right away. With mobile data, you can send support, check on your transactions, and get updates wherever you are no need to stop what you're doing.

Peopleware Requirements

Building, running, and keeping the system working takes a team of skilled, experienced people. Peopleware spells out exactly who does what both on the tech side and in management at every stage of the system's life.

Project Manager

The Project Manager does all the heavy lifting when it comes to system development. They lay out the timeline, establish milestones, and determine who needs to do what and when. Their role in the system development process is to ensure that everything remains on schedule and that the system gets developed the way it should. They also ensure that everyone remains updated, from the team members to everyone else, so that nothing gets missed.

System Developer

The System Developer does it all for system architecture. They design and build everything from scratch blockchain framework, backend services, databases, analytics, you name it. They're the one making sure it's all secure, runs smoothly as it grows, and just works the way it should.

Mobile Application Developer

The Mobile Application Developer is responsible for designing, coding, and testing the systems mobile application. This includes ensuring compatibility with Android and iOS platforms, implementing user-friendly interfaces, and integrating blockchain-based donation features. The developer collaborates closely with the backend team to ensure seamless data synchronization.

Web Developer

The Web Developer is responsible for developing and maintaining the web-based version of the system. This includes creating responsive user interfaces, implementing administrative dashboards, and ensuring secure access to system features. The Web Developer works with other team members to ensure proper system integration.

UI/UX Developer

The UI/UX Developer is responsible for designing intuitive, accessible, and visually appealing user interfaces for both the mobile and web applications. This role includes conducting user research, creating wireframes and prototypes, and ensuring a positive user experience for donors, administrators, and NGO staff.

Blockchain Specialist

The Blockchain Specialist is responsible for implementing and managing the blockchain components of the system. This includes smart contract development, transaction validation, and ensuring data integrity and transparency within the donation process.

Data Analyst

The Data Analyst is responsible for implementing descriptive and prescriptive analytics within the system. This role involves analyzing donation trends, generating insights, and supporting decision-making processes for NGOs through data visualization and reports.

3.2 PROJECT TECHNICAL DESCRIPTION

The ReliefLink system is designed to support transparent and accountable donation management through a mobile application for users and a web-based platform for administrators. This section presents the system's technical description by outlining its core functionalities, system components, and integrated technologies to provide a clear understanding of how the system operates during development, testing, and deployment.

The system follows a client-server architecture where data processing, storage, and validation are handled through a centralized backend. The mobile application serves as the primary platform for user interaction, while the web application is dedicated to administrative functions such as monitoring transactions, verifying records, and managing reports. To ensure data integrity and transparency, blockchain technology is integrated into critical system modules, particularly in donation recording and fund distribution tracking.

Special Features of the System

The system utilizes modern web and mobile development technologies, blockchain solutions, and data analytics tools to enhance transparency, security, and decision-making in donation management. Designed to promote trust and accountability, the platform combines secure transaction recording with analytical insights to support efficient system operations.

Blockchain technology is integrated into the donation and fund utilization modules to securely record cash-based transactions in an immutable ledger. Each donation, verification entry, and fund release record is permanently stored once validated, preventing unauthorized modification or data tampering. This ensures a transparent and auditable history of transactions that can be reviewed by administrators at any time.

Descriptive analytics are implemented within the administrative dashboard to process and analyze historical donation and distribution data. Descriptive analytics summarizes donation patterns, fund allocation frequency, and reporting trends through visual dashboards, allowing administrators to easily interpret system activity. In addition, the system includes real-time notification services that provide timely updates regarding donation status, verification results, and system activities. These notifications help maintain continuous communication between system components and administrators, ensuring efficient workflow management and prompt response to system events.

System Architecture

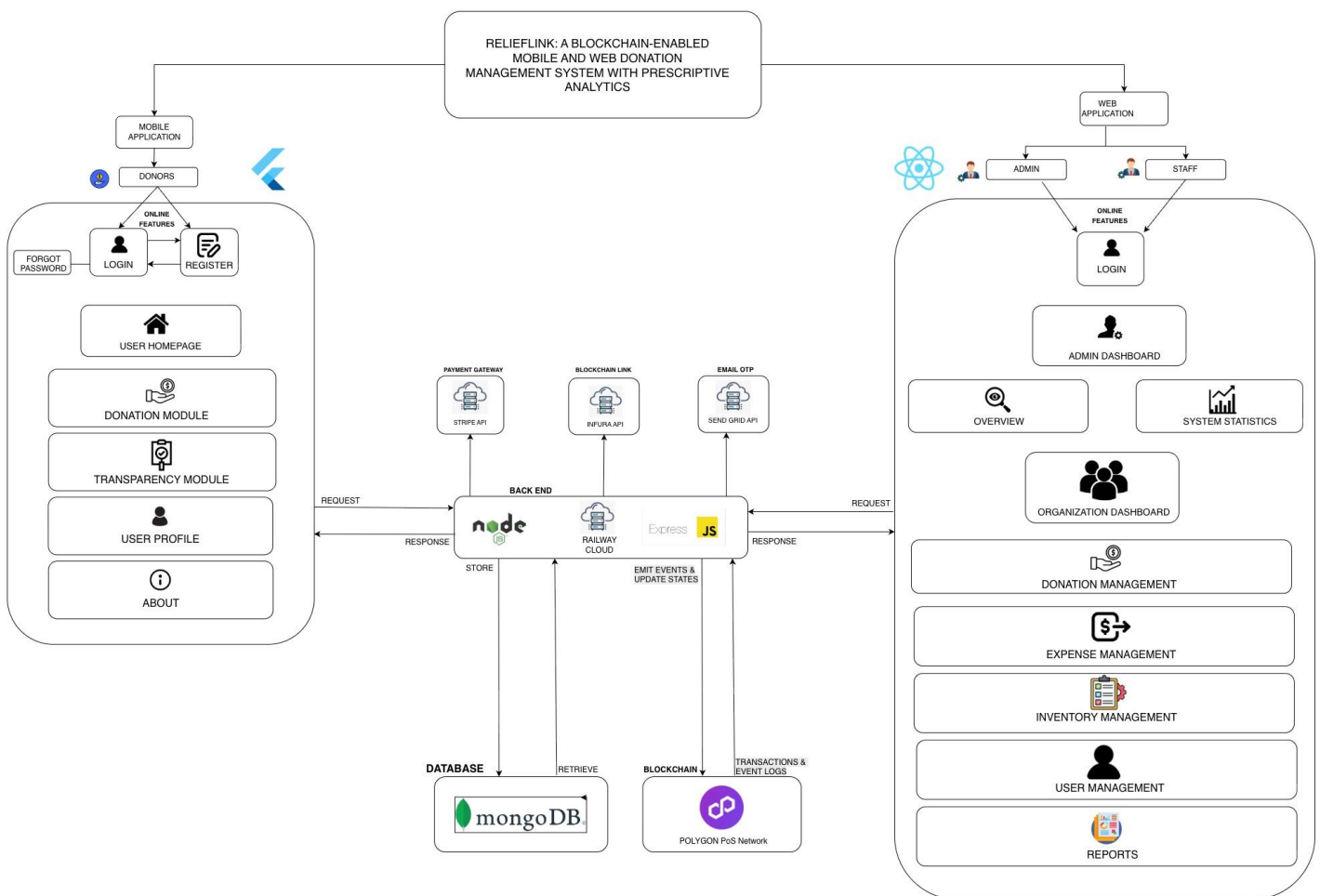


Figure 1 Overall System Architecture

ReliefLink is a comprehensive, blockchain-enabled donation management platform designed to bring transparency and data-driven efficiency to charitable giving. The user journey begins at the login and registration portal, which serves both the mobile application for donors and the web dashboard for administrators. Users, including donors, admin, and staff, authenticate through a secure system built with Node.js and Express.js, which validates credentials against the MongoDB database. If a user forgets their password, a "Forgot Password" feature is triggered, initiating a secure process where the backend sends an email containing a one-time password (OTP) to facilitate a verified reset.

Once logged in, donors access the Flutter-based mobile application, which features a personalized user homepage. From here, they can engage with the core Donation Module to browse causes and contribute funds, with all payments securely processed through the integrated Stripe API payment gateway. A key component of the mobile experience is the Transparency Module, which allows donors to track the usage of their contributions via immutable records stored on the blockchain, fostering trust. Users can also manage their personal details in the User Profile section and learn more about the platform in the About page.

For administrative operations, authorized staff and administrators log into a separate React-based web dashboard. This dashboard provides a comprehensive overview and system statistics, acting as a central command center. It includes several management modules: an Organization Dashboard for a high-level view, Donation Management to track incoming funds, Expense Management to log and justify operational costs, Inventory Management for relief goods, and User Management to oversee all platform accounts. Furthermore, the system generates detailed Reports powered by prescriptive analytics, offering actionable insights for optimal resource allocation.

The entire ecosystem is powered by a robust backend. The Node.js and Express.js server handles all application logic, API requests, and communication between the frontend clients and the MongoDB database, which stores all critical data such as user profiles, transactions, and inventory records. This backend also manages third-party integrations, including the Stripe API for financial processing and the email service for dispatching OTPs and notifications. Ultimately, the blockchain layer is woven throughout this architecture, providing an unchangeable ledger for all donation and expense transactions, which is the foundation for the system's prescriptive

analytics and its core promise of verifiable transparency from the point of login to every backend operation.

Table 4 System Modules of the Project

Module No.	Module Name	Description
[SYSTEM/APPLICATION] Web		
M1	Login Module	Allows administrators and staff to securely log in to the web application.
M2	OTP Verification Module	Provides additional security through email-based OTP verification during login.
M3	Admin Dashboard Module	Displays an overview of the system including system statistics and summaries.
M4	Donation Management Module – View Donations	Allows administrators and staff to view all recorded donations.
M5	Donation Management Module – Verify Donations	Enables verification and validation of submitted donation records.
M6	Donation Management Module – Record Cash Donations	Allows administrators to manually record cash donations.
M7	Expense Management Module – Record Fund Usage	Records how donated funds are utilized by the organization.
M8	Expense Management Module – Upload Expense Receipt	Allows uploading of receipts as proof of fund utilization.
M9	Expense Management Module – Link Expense to Donation Pool	Links recorded expenses to specific donation funds for accountability.
M10	Inventory Management Module – Food Inventory Management	Manages records of food supplies and inventory.
M11	Inventory Management Module – Distribution Records	Records food distribution activities to beneficiaries.
M12	User Management Module – Manage Users	Allows administrators to create, edit, and deactivate system users.
M13	User Management Module – Role Assignments	Assigns system roles and permissions to users.
M14	Reports Module – Donation Reports	Generates reports related to donation transactions.
M15	Reports Module – Expense Reports	Generates reports related to fund usage and expenses.
M16	Reports Module – Donation Logs	Displays immutable logs of donation activities for auditing purposes.
[SYSTEM/APPLICATION] Mobile		
M1	Registration Module	Allows donors to register an account in the system using their personal information.
M2	Login Module	Enables registered donors to securely log in to the mobile application.
M3	Forgot Password (OTP) Module	Allows users to reset their password through email-based OTP verification.

M4	Donation Module – Make Donations	Enables donors to submit cash-based donation details through the mobile application.
M5	Donation Module – Donation History	Displays the complete list of donations made by the donor.
M6	Donation Module – Track Donation Status	Allows donors to monitor the status of their submitted donations.
M7	Donation Summary Module	Displays a summarized view of the donor's donation records.
M8	Reports Viewer Module (Read-Only)	Allows donors to view donation reports in read-only mode.
M9	User Profile Module	Allows donors to view and manage their profile information.
M10	About Module	Displays information about the ReliefLink system and the organization.

CHAPTER 4

METHODOLOGY, SYSTEM PROCESSES AND DESIGNS

This chapter presents the methodology, requirement analysis and documentation, the design of the system using different models and the development of the proposed system application.

4.1 REQUIREMENTS ANALYSIS AND DOCUMENTATION

Research Methodology Used

The research used descriptive methods to establish support for the ReliefLink system which functions as a donation management system with blockchain technology and dual web and mobile interfaces. The research method of descriptive research studies existing situations through identification and analysis of current conditions and existing challenges and operational methods. These exist in the research field.

The research method allowed the authors to perform an in-depth analysis of how NGOs handle donations while they studied what donors expect and what technology systems exist for donation management. The research process involved document analysis and literature review and donation platform assessment, which revealed standard problems that included donor information concealment and donation monitoring system flaws and report delivery problems and restricted data analytics application for organizational choices.

The research team analyzed the collected data to identify system requirements, which included both functional and non-functional elements. The research block analysis established the basis for creating a blockchain solution, which enables organizations to achieve better visibility and self-governance, and data-based decision-making abilities. The descriptive research approach ensured that the proposed system aligns with real-world NGO operations and effectively addresses the identified challenges.

Project Development Methodology

The project development methodology defines the structured framework used for the design, implementation, and evaluation of ReliefLink: A Blockchain-Enabled Mobile and Web Donation Management System with Prescriptive Analytics. After careful analysis of the system requirements, technological complexity, and operational environment, the Agile Scrum Software Development Methodology was selected as the most appropriate approach for this project.

Agile Scrum is an iterative and incremental development framework that emphasizes flexibility, stakeholder collaboration, and continuous system improvement. Unlike traditional waterfall models that follow a rigid sequence of phases, Agile Scrum supports adaptive planning and evolutionary development through short development cycles known as sprints, each delivering functional and testable system components [6], [7].

Agile Scrum methodology is particularly suitable for mobile and web-based systems that integrate emerging technologies such as blockchain and data analytics. Blockchain applications require continuous validation of transaction integrity, immutability, and security, which can be effectively achieved through Agile's incremental development and continuous integration practices [3]. Furthermore, analytics-driven systems benefit from iterative model development and evaluation to ensure accurate and reliable decision recommendations [4]. Agile also supports compliance with international software quality standards such as ISO/IEC 25010 by embedding testing and quality assurance throughout the development lifecycle, ensuring functionality, reliability, usability, security, and maintainability of the system [5].

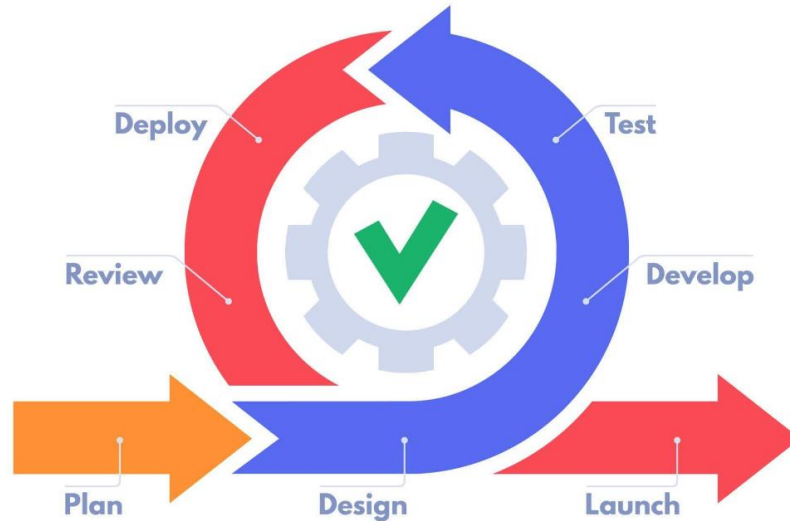


Figure 2 Project Development Methodology

The development of ReliefLink: A Blockchain-Enabled Mobile and Web Donation Management System with Prescriptive Analytics follows an Agile Scrum-based development approach structured into multiple iterative sprints, each with a duration of two weeks. The entire development cycle is planned to be completed within a three-month timeframe. Each sprint focuses on the implementation of specific system components such as donor registration, blockchain-based donation recording, receipt verification, prescriptive analytics processing, and administrative reporting. At the end of every sprint, system testing and stakeholder evaluation are conducted to ensure continuous system improvement and early detection of technical and operational issues [6], [7].

The Agile Scrum development approach for ReliefLink is guided by the four core values of the Agile Manifesto [6]:

- Individuals and interactions over processes and tools

The development process prioritizes effective communication and collaboration among the development team, system administrators of the KEH Family Foundation, and project stakeholders. Regular sprint meetings, demonstrations, and feedback sessions ensure that system requirements are clearly understood and accurately implemented.

- Working software over comprehensive documentation

While necessary technical and user documentation is maintained, the primary focus of the development process is the delivery of functional and testable

system features. Priority is given to core functionalities such as blockchain transaction logging, donation tracking, receipt verification, and prescriptive analytics recommendations.

- Customer collaboration over contract negotiation

Continuous involvement of foundation administrators and donors is integrated throughout the development process. Their participation in sprint reviews and system evaluations ensures that ReliefLink addresses actual operational needs and improves transparency and accountability in donation management

- Responding to change over following a plan

The development approach embraces changes in system requirements, security policies, and analytics models to ensure that the system remains aligned with evolving organizational processes, donor expectations, and technological advancements in blockchain and data analytics.

Through this Agile Scrum-driven development approach, ReliefLink is implemented as a flexible, scalable, and user-centered donation management platform that continuously evolves based on real-world feedback and operational demands [7], [8].

Agile Development Phases

The ReliefLink development process follows an iterative Agile Scrum approach, structured into seven detailed phases. Each phase is executed in two-week sprints, which allows incremental development, continuous testing, and iterative refinement based on stakeholder feedback [6], [7]. This ensures that the system remains flexible, reliable, and aligned with organizational goals while integrating blockchain technology and prescriptive analytics [8].

Phase 1: Plan

During the planning phase, system requirements are gathered from the administrators and donors of the KEH Family Foundation to identify the operational needs of the donation management process. Functional requirements such as donor registration, secure login, donation processing, blockchain-based transaction recording, receipt verification, donation tracking, analytics reporting, and administrative monitoring are defined. Non-functional requirements including system security,

performance, availability, and scalability are also identified. The project scope, objectives, and user roles are established, and a feasibility study is conducted to assess technical and operational viability. Risk analysis is performed to identify potential issues related to data security, blockchain integration, and system reliability [6], [7].

Phase 2: Design

The design phase translates system requirements into technical and functional specifications for both the mobile and web platforms. The overall system architecture is defined, including the integration of the blockchain ledger for immutable donation records, the database for storing user and transaction data, and the prescriptive analytics engine for decision support. User interface and user experience designs are created for donor registration, donation submission, transaction tracking, and administrative dashboards. Security mechanisms such as authentication, encryption, and role-based access control are incorporated into the system design. Data flow diagrams, entity-relationship diagrams, and system architecture diagrams are prepared to visualize system interactions [7], [8].

Phase 3: Develop

In the development phase, the ReliefLink system modules are implemented incrementally based on sprint priorities. Core modules include the donor registration and authentication module, the donation processing module with blockchain transaction logging, the receipt upload and verification module, the donation history and tracking module, the prescriptive analytics engine, and the administrative reporting dashboard. Continuous integration is applied to ensure seamless interaction between the mobile application, web platform, blockchain network, and database. The blockchain component ensures transaction immutability and transparency, while the analytics module provides data-driven insights for fund allocation and decision-making [7], [8].

Phase 4: Test

The testing phase evaluates the functionality, security, performance, and usability of the ReliefLink system. Unit testing is conducted for individual modules, while integration testing verifies communication between the mobile app, web platform, blockchain ledger, and analytics services. Security testing is performed to validate encryption mechanisms, authentication processes, and blockchain integrity.

Performance testing ensures that the system can handle concurrent users and large volumes of donation transactions. User Acceptance Testing (UAT) is conducted with administrators and donors to verify that the system meets operational requirements. The system is evaluated based on ISO/IEC 25010 quality standards, including functionality, reliability, usability, security, and maintainability [10], [11].

Phase 5: Deploy

During deployment, the ReliefLink web platform is hosted on secure cloud servers, and the mobile application is prepared for Android and iOS distribution. Blockchain nodes, database servers, and analytics services are configured to support real-time donation processing and reporting. Backup and recovery mechanisms are implemented to ensure data protection and system availability. Final integration and validation tests are conducted to confirm that all modules are fully operational prior to public release [8], [5].

Phase 6: Review

The review phase focuses on evaluating system performance, user experience, and analytics accuracy after deployment. Feedback is collected from donors and administrators regarding system usability, transparency, and operational efficiency. Blockchain transaction logs and system performance metrics are analyzed to detect anomalies and optimize system operations. Enhancements are applied to improve interface usability, processing speed, security controls, and analytics recommendations [9], [11].

Phase 7: Launch

The launch phase marks the official release of the ReliefLink system for operational use. The final system version is deployed to production servers, and training sessions are conducted for foundation administrators and staff. User manuals and technical documentation are distributed to support system adoption. Post-launch monitoring is implemented to ensure continuous system availability, security, and performance. User feedback is continuously gathered to support future system maintenance and enhancement.

Table 5 Project Development Phases, Activities and Deliverables

Phases	Activities	Deliverables
Plan	• Gather functional and non-functional requirements from stakeholders, including administrators and donors. • Define project scope, objectives, and user roles. • Conduct feasibility study to assess technical, operational, and financial viability. • Perform risk assessment to anticipate challenges such as data breaches or blockchain integration issues.	• Project plan • Requirements specification document • User stories • Use-case diagrams
Design	• Design system architecture for web and mobile platforms. • Develop database schema and blockchain ledger structure. • Design prescriptive analytics engine for optimized fund allocation. • Prepare UI/UX wireframes for accessibility and usability. • Incorporate security measures including authentication, encryption, and role-based access control. • Create ER diagrams, data flow diagrams, and architecture diagrams to visualize data movement.	• System design document • Wireframes and mockups • Architecture diagrams • Security design specification
Develop	• Implement donor registration module.	• Functional system modules

	• Implement donation processing module with blockchain logging. • Develop receipt verification module. • Implement analytics engine for fund allocation. • Develop administrative reporting module. • Perform continuous integration and code review.	• Source code repository • Integrated blockchain module • Analytics engine module
Test	• Conduct unit testing for individual modules. • Perform integration testing across modules. • Conduct security and performance testing. • Conduct usability testing for administrators and donors. • Execute User Acceptance Testing (UAT). • Evaluate system based on ISO/IEC 25010 standards for functionality, reliability, usability, security, and maintainability.	• Test cases • Test reports • UAT feedback • Evaluation results
Deploy	• Deploy web platform on secure servers. • Package and distribute mobile applications for Android and iOS. • Configure blockchain nodes and database servers.	• Deployed web and mobile system • Configured blockchain network • Monitoring and backup setup

	• Implement system monitoring and backup mechanisms. • Conduct final integration verification to ensure all modules function correctly.	
Review	• Collect feedback from stakeholders regarding usability and functionality. • Review analytics outputs to verify accuracy of fund allocation recommendations. • Monitor system logs and blockchain transactions for anomalies. • Identify areas for optimization in workflows, usability, or security. • Plan iterative refinements for subsequent development cycles.	• Review report • Updated system version • Validated analytics reports
Launch	• Conduct final deployment of ReliefLink system. • Provide training sessions for administrators and staff. • Distribute user manuals and technical documentation. • Monitor system performance post-launch. • Collect post-launch feedback to guide maintenance and future upgrades.	• Final system release • User manuals • Technical documentation • Post-launch performance reports

Functional Requirements

Table 6 Functional Requirements of the Project

FR No.	Module No.	Function Name(s)	Description
FR-01	M1	User Registration and Login	The system allows users to create accounts and log in securely.
FR-02	M1	User Profile Management	The system will allow users to view and update personal information.
FR-03	M2	NGO Registration and Verification	The system will allow NGOs to register and be verified by the administrator.
FR-04	M3	Donation Campaign Management	The system will allow NGOs to create, update, and manage donation campaigns.
FR-05	M3	Donation Processing	The system will allow donors to donate through supported payment methods.
FR-06	M4	Blockchain Transaction Recording	The system will record donation transactions on the blockchain ledger.
FR-07	M5	Donation Tracking	The system will allow users to track donation history and transaction status.
FR-08	M6	Descriptive Analytics	The system prepares summary reports and visual analytics.
FR-09	M6	Prescriptive Analytics	The system will provide recommendations to improve campaign performance.
FR-10	M7	Notification System	The system users will be notified of transactions and system updates.

Non-Functional Requirements

Table 7 Non-Functional Requirements of the Project

Non-Functional Criteria	Description
Functionality	The system should: 1. Provide accurate and complete donation management features.

	2. Support essential operations such as donations, user management, and reporting. The system is constrained to: 1. Features as implemented in the current system design. 2. External integrations dependent on available APIs.
Performance	The system should: 1. Respond efficiently under normal and peak usage. 2. Process requests and transactions quickly without delays. The system is constrained to: 1. Server capacity and network bandwidth. 2. Hardware limitations of client devices.
Scalability	The system should: 1. Handle increasing numbers of users and donation transactions. 2. Support growth without degrading performance. The system is constrained to: 1. Database and server architecture limits. 2. Resource allocation and storage capacity.
Portability	The system should: 1. Be accessible online and on mobile devices. 2. Function correctly across multiple platforms. The system is constrained to: 1. Supported browsers and mobile operating systems. 2. Device hardware limitations.
Compatibility	The system should: 1. Work correctly across different browsers and devices.

	2. Support modern web standards and device resolutions. The system is constrained to: 1. Common modern browsers and devices only. 2. Certain legacy systems may not be fully supported.
Reliability	The system should: 1. Ensure consistent operation and accurate data processing. 2. Recover gracefully from minor failures. The system is constrained to: 1. Network stability and server uptime. 2. External service dependencies.
Availability	The system should: 1. Remain online with minimal downtime. 2. Ensure critical services are accessible at all times. The system is constrained to: 1. Scheduled maintenance periods. 2. Unexpected technical issues.
Maintainability	The system should: 1. Use a modular architecture for easier updates. 2. Include documentation for system components and APIs. 3. Implement consistent coding standards across the codebase. The system is constrained to: 1. Maintenance performed only by the development team. 2. Limited knowledge transfer to new developers or end users.
Security	The system should:

	1. Protect user data and transactions with encryption and blockchain. 2. Prevent unauthorized access or tampering. The system is constrained to: 1. Known vulnerabilities in encryption or blockchain protocols. 2. Security policies and practices followed by the development team.
Usability	The system should: 1. Provide an intuitive and user-friendly interface. 2. Make navigation and key functions simple for all users. The system is constrained to: 1. Design framework limitations. 2. Consistency of interface elements across the system.

4.2 DESIGN OF SOFTWARE, SYSTEMS, PRODUCTS AND/OR PROCESSES

Use Case Diagram

In this section, present and discuss the Use Case Diagram of the System (MWA – both web and mobile system; MAA – either web, desktop or mobile system) (Use Style: Capstone Content – Body Text). Note: Diagrams symbols, flows and labels must be readable. See example below for reference.

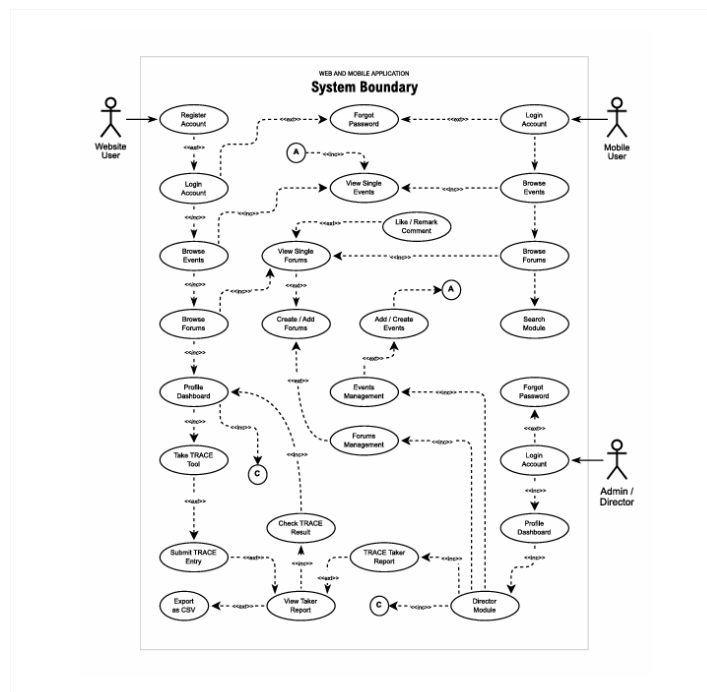


Figure 3 Use Case Diagram of the System

Describe briefly the Use Case diagram in this section.

Table 8 Use Case Narratives of the System

Use Case Name	Admin Login
Actor(s)	Administrator
Description	This use case allows the administrator to securely access the web-based administrative system using registered credentials. The system verifies the administrator's identity before granting access to administrative features.
Pre-condition	The administrator has a registered admin account.
Post-condition	The administrator is successfully logged in and redirected to the admin dashboard.
Main Scenario	• The administrator opens the web-based admin system. • The administrator enters valid login credentials. • The system verifies the credentials. • The administrator is redirected to the admin dashboard.
Alternative Scenario	If invalid credentials are entered, the system displays an error message and denies access.

Use Case Name	Manage System Content and Advertisements
Actor(s)	Administrator
Description	This use case allows the administrator to manage website content that serves as an advertisement and information portal for the mobile application. The administrator can update system descriptions, features, announcements, and promotional materials encouraging users to download the mobile app.
Pre-condition	The administrator is logged in to the web-based admin system.
Post-condition	Updated system information and advertisements are published and visible on the website and mobile application.
Main Scenario	• The administrator accesses the content management section.

	• The administrator creates or edits system information, announcements, or advertisements. • The administrator saves the changes. • The system updates the content and confirms successful publishing
Alternative Scenario	If required content fields are incomplete, the system displays an error message and prevents saving.

Use Case Name	Monitor System Records
Actor(s)	Administrator
Description	This use case enables the administrator to monitor donation records, verification logs, and overall system activity generated from the mobile application for transparency and auditing purposes.
Pre-condition	The administrator is logged in to the admin dashboard.
Post-condition	The administrator successfully views updated system records and reports.
Main Scenario	• The administrator selects the system monitoring option. • The system retrieves records from the database. • The administrator reviews reports and transaction logs.
Alternative Scenario	If the system fails to load records, an error message is displayed.

Use Case Name	User Registration
Actor(s)	User
Description	This use case allows users to create an account in the mobile application by providing the required personal information.
Pre-condition	The user has installed the mobile application.

Post-condition	The user account is successfully created and stored in the system.
Main Scenario	• The user opens the mobile application. • The user selects the “Register” option. • The user fills out the registration form. • The system validates the information. • The system creates the user account and displays a success message.
Alternative Scenario	If invalid or incomplete data is submitted, the system displays an error message.

Use Case Name	User Login
Actor(s)	User
Description	This use case allows registered users to log in to the mobile application to access system features.
Pre-condition	The user has a registered account.
Post-condition	The user is logged in and redirected to the home screen.
Main Scenario	• The user enters login credentials. • The system verifies the credentials. • The user is redirected to the home screen.
Alternative Scenario	If credentials are invalid, an error message is shown.

Use Case Name	View Donation Records and Status
Actor(s)	User
Description	This use case allows users to view real-time donation records, transaction history, and fund distribution status stored in the blockchain-enabled system.

Pre-condition	The user is logged in to the mobile application.
Post-condition	Donation records and statuses are displayed to the user.
Main Scenario	• The user selects the donation records option. • The system retrieves verified donation data. • The user views transaction details and status updates.
Alternative Scenario	If data cannot be retrieved, the system displays a loading or error message.

Use Case Name	Upload Receipt and Verification Documents
Actor(s)	User
Description	This use case enables users to upload receipts and supporting documents for donation verification, ensuring transparency and accountability.
Pre-condition	The user is logged in and has a completed donation transaction.
Post-condition	The uploaded documents are stored and submitted for verification.
Main Scenario	• The user selects the upload verification option. • The user uploads receipt or document files. • The system validates and stores the files. • The system confirms successful submission.
Alternative Scenario	If files are missing or invalid, the system displays an error message.

Functional Decomposition Diagrams

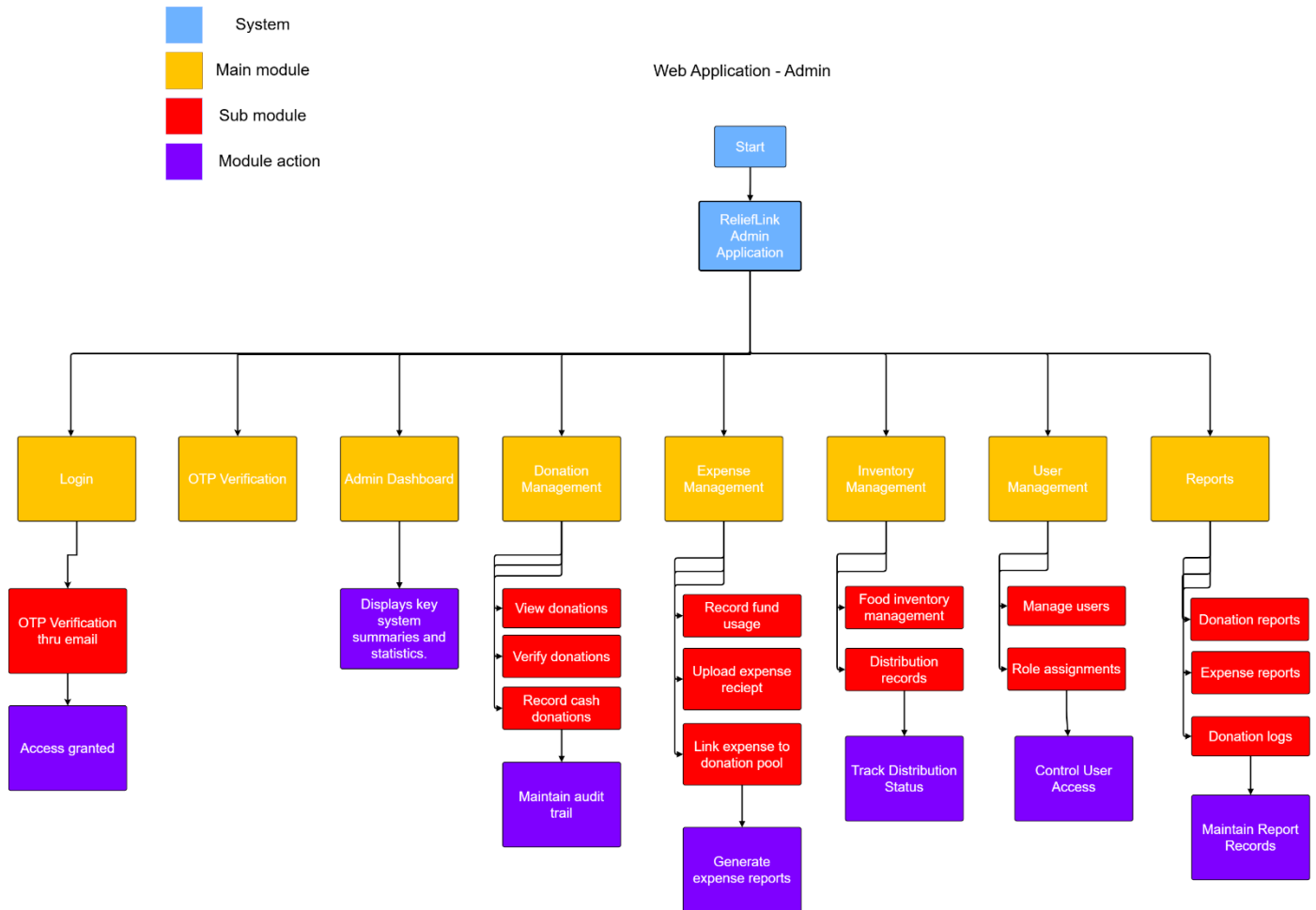


Figure 4 Functional Decomposition Diagram of the Web Application - Admin

ReliefLink Admin Application is the main administration system in the overall system. It is meant to facilitate effective donation tracking, financial management, inventory management, and management of users. Additionally, the application provides administrators with secure and necessary tools for managing operations.

Login Module is the security gateway for the system. The administrators must authenticate their credentials using the secure login system provided with email OTP. The two-factor system ensures that only authorized people have access to the administrative system.

Admin Dashboard Module serves as the application's central analytics hub. It displays key system summaries and statistics, enabling administrators to view in real time donations, expenses, inventory levels, and overall system performance. This

module supports descriptive analytics for the purpose of assisting administrators in monitoring the status of operations and trends.

Donation Management Module is designed for managing all types of activities related to donations. The administrators have access to view and confirm all types of donations, which ensures that all types of donations are authentic. Additionally, this module will enable administrators to enter cash donations that were made offline. To ensure transparency, all types of donation transaction details will be recorded, along with an audit trail.

Expense Management Module offers functionalities that enable administrators to record expenditures, upload receipts, and assign expenditures to the donation fund to ensure that all expenditures are tracked and recorded. The administrators can also produce reports to aid analysis and tracking in case of an audit.

Inventory Management Module is a tool that supports the monitoring and control of food, supply, and other resources. With the system's food inventory management, administrators can monitor the current levels of the available supply, while the distribution data shows the allocation of the supplies. This tool also allows administrators to have information on the distribution status of the items, which offers a mechanism that prevents misallocation of the resources.

User Management Module enables the administration of users on the system. An administrator can conduct the administration of users by using the role-assignment functions, along with other functions. This enables the administrator to limit access by assigned users based on permission provided.

Reports Module is responsible for organizing system information into a formatted report that can then be viewed and assessed. System administrators can produce reports on donations and expenses, giving a historical account of all system processes. The module also guarantees the preservation of records of reports produced and stored within the system.

This hierarchical functional structure demonstrates how the ReliefLink Admin Application integrates secure access, financial management, inventory monitoring, user control, and reporting capabilities. The system supports efficient administrative operations while promoting transparency, accountability, and effective governance within the system platform.

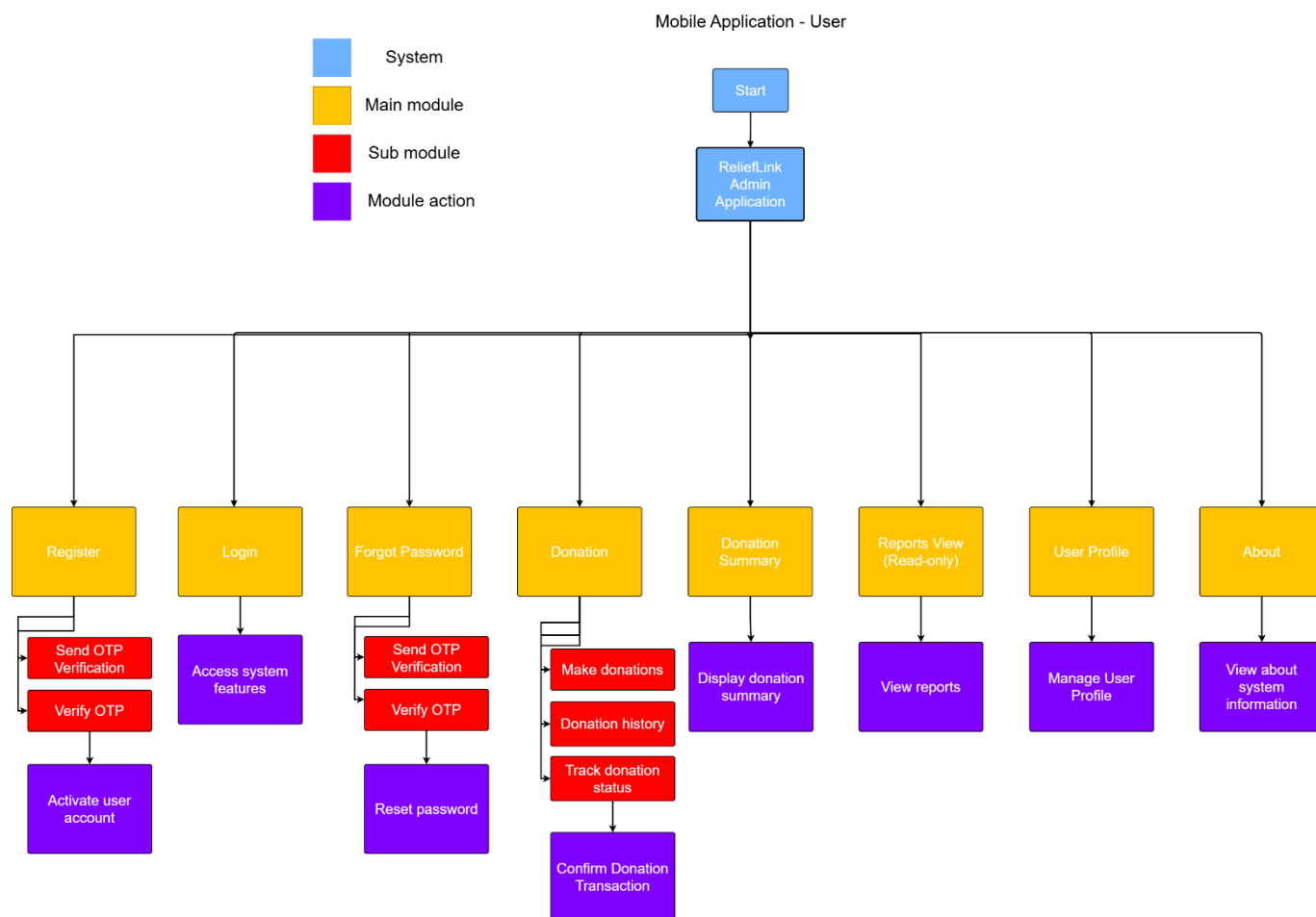


Figure 5 Functional Decomposition Diagram for Mobile Application

In this paragraph, briefly describe the illustration on various functions that user can perform.

Process Flow Diagrams



Figure 6 Process Flow – Module Name

Present in this section the process flow of the system (users vis-à-vis system) using Activity Diagram with Swim Lanes (MWA – both web and mobile system; MAA – either web or mobile system, if applicable otherwise MAA required designs). Describe each illustration explaining the various processes involved in the specific module (Use Style: Capstone Content – Body Text). Note: Diagrams symbols, flows and labels must be readable.



Figure 7 Process Flow – Module Name

Present in this section all the process flow of the system (users vis-à-vis system modules) using Activity Diagram with Swim Lanes (MWA – both web and mobile system; MAA – either web or mobile system, if applicable otherwise MAA required designs). Describe each illustration explaining the various processes involved in the specific module (Use Style: Capstone Content – Body Text). Note: Diagrams symbols, flows and labels must be readable.

4.3 DESCRIPTION OF THE SYSTEM

In this section of your capstone manuscript, you should include illustration of all major screen designs (MWA – both Web and Mobile Systems; MAA – Storyboard for Film or Animation; MAA System Development – either Web or Mobile Systems) and should contain a brief description of its functions and features. Detailed screen designs (UI) should be included in the Appendix section of your manuscript.



Figure 8 Web Application System

Describe each individual screen designs (module) by explaining the main functions and features. Repeat for other major screen designs (Use Style: Capstone Content – Body Text).



Figure 9 Mobile Application System

Describe each individual screen designs (module) by explaining the main functions and features. Repeat for other major screen designs (Use Style: Capstone Content – Body Text).

REFERENCES

In this section, use appropriate referencing tools such as Mendeley or Zotero. Use IEEE refencing styles accordingly (refer to this link <https://iee-dataport.org/sites/default/files/analysis/27/IEEE%20Citation%20Guidelines.pdf> as a guide).

[1] A. D. Ariffin, N. A. Ghani, and N. A. M. Isa, "Blockchain Based Donation Management in Disaster Response," J. Inf. Technol. Impact, vol. 6, no. 1, p. 2257, 2021. doi: 10.35970/jinita.v6i1.2257.

[2] S. R. K. R. A. Uddin, M. A. Hossain, and M. A. Rahman, "Donation Transparency Through Blockchain Technology: A Bibliometric Analysis," Int. J. Adv. Comput. Sci. Appl., vol. 12, no. 5, pp. 82–91, 2021. doi: 10.33102/82gb3a38.

[3] P. Kumar, M. S. Hossain, and R. Sharma, "A Framework to Make Charity Collection Transparent and Auditable Using Blockchain Technology," Comput. Electr. Eng., vol. 87, p. 106588, 2020. doi: 10.1016/j.compeleceng.2020.106588.

[4] M. Alharby and A. van Moorsel, "Blockchain-based Donations Traceability Framework," J. King Saud Univ. Comput. Inf. Sci., vol. 34, no. 9, pp. 5690–5704, 2022. doi: 10.1016/j.jksuci.2022.09.021.

[5] Y. Zhang, J. Wu, and X. Li, "Blockchain-Empowered Decentralized Philanthropic Charity for Social Good," Sustainability, vol. 16, no. 1, p. 210, 2024. doi: 10.3390/su16010210.

APPENDICES

APPENDIX A.

LETTERS AND

DOCUMENTATIONS

In this section of your Appendix, scanned clear copy and/or actual document must be included. Note: Documents that should have signatory fields must be completely signed (wet mark is a MUST). Document should be available are as follow(s):

- *[Forms] Request Letter for Client Interview with Non-Disclosure Agreement*
- *[Forms] Request Letter for Capstone Adviser (Letter and Advising Contract)*
- *[Forms] Capstone Consultation Logs (Student's Copy Capstone 1 – minimum 10)*
- *[Forms] System Development Progress Report (All Report from Capstone 1)*

APPENDIX B.

TRANSCRIPT OF

INTERVIEWS AND

PHOTO DOCUMENTATION

In this section of your Appendix, scanned clear copy and/or actual document must be included. Note: Documents that should have signatory fields must be completely signed (wet mark is a MUST). Document should be available are as follow(s):

- *Transcript of Interview (All conducted interviews and visits)*
- *Photo Documentation (Proof of Client Interview, Visitation and Consultation)*

APPENDIX C. PANEL'S LIST OF RECOMMENDATIONS AND COMPLIANCES

In this section of your Appendix, scanned clear copy and/or actual document must be included. Note: Documents that should have signatory fields must be completely signed (wet mark is a MUST). Document should be available are as follow(s):

- *Capstone Project Proposal*
 - *Panel's Recommendation Approval and Compliance*
- *Capstone Project 1 (should be available for manuscript deliverable requirement)*

- *Panel's Recommendation Approval and Compliance*